

Unit-3

Car on fire

- 1) Electrical design — S.C. / leakage etc.
- 2) Thermal design — Temp. / places matter
- 3) Life — corrode ^{cells} should not move, terrain defines life. In big cities lot of jerking due to break
- 4) Durability — should run for its given life
- 5) Safety — Utmost imp., many factors.
- 6) Performance — Resis. cause problem in currents, materials used should ~~have~~ be proper for optimum performance.

Battery pack design / development process

Electrical Design:— Harness, BMS, Busbars, Connectors etc.

Thermal management, — required/ betⁿ 15 to 25°C

Mechanical design

BMS ;— Heart of Battery

- 1) Current flow during charge / discharge
- 2) communicates with charger
- 3) ————— Driver
- 4) It controls cell temp by cutting off the supply
- 5) If current asked by (m) is above (way above) its limit BMS will cut off the Supply.

Importance of

Series | combination
of cells.

1) Electrical design : —

- 1) Capacity, Voltage & current
- 2) High voltage isolation — High capacity bat
- 3) Short circuit Scenarios → +ve touches -ve
- 4) Efficient power delivery

↑ leakage
applied
DC voltage is
Safe only
60 to 70V

2) Thermal Design : —

- 1) Improve pack efficiency.
- 2) Mitigate thermal accidents
- 3) Increase cell/pack life

every 10°C ↑ brings life ↓
→ running in: certain limit.

3) Mechanical design.

- 1) Safe structure for extreme conditions
- 2) Cost, productivity and reliability
- 3) Ease of assembly & service
- 4) Aesthetics, compactness & lightweight.

4) BMS Design

- 1) Maintain cell/pack operation limits
- 2) Prevent safety concerning events
- 3) Control Thermal systems
- 4) Communication & diagnostics.

Stages of Battery Pack Design Pic

Pic

Building Battery pack

2 Wheeler EV for 75km which consumes 16.5 kWh/km.
on average from battery pack to run.

Considering depth of discharge (DoD) to be 80%,
design battery pack of nominal operating
voltage 60V.

Power consumed by auxiliary system is 2 Wh/km

Cell specifications as follows:-

$$\frac{1.7 \text{ kWh}}{60 \text{ V}}$$

$$= 27.28 \text{ Ah}$$

15 Ah cell
2 cells in 1

2P16S

In 1 row 2P8S

Total no. of cells 32, Rows-2

No. of cells in 1 row = 16

Total weight of cells in 1 row: $32 * 0.32 = 10.24 \text{ kg}$.

$$18.5 \text{ Wh/km}$$

$$75 \times 18.5 \text{ km} = 1.4 \text{ kWh}$$

$$0.8 \rightarrow \text{DoD} = 1.7 \text{ kWh}$$

$$\frac{60 \text{ V}}{3.7 \text{ V}}$$

$$16.2$$

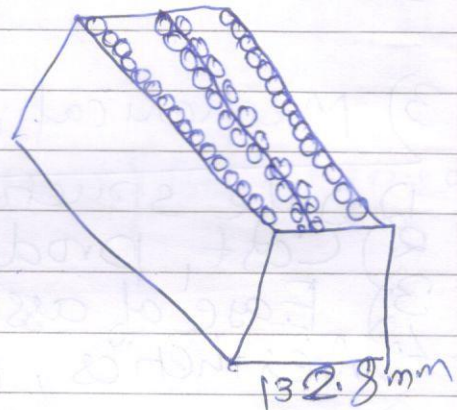
take 16 in series.

nominal voltage

3.1V

4.2V

Go 50V to 70V whole battery



~~stress & strain theory~~

Types of loading.

Base plate requirement take Cell load

- without bending
 - without breaking
- } withstanding all jerks & vibrations while driving.

For Mech Design :- Important Consideration

- 1) Material Selection $\rightarrow$ Ashby Methodology.
- 2) Base plate design for individual cell.
- 3) Cell movement constraint & control.
- 4) Uniform pressure over cell surfaces.
- 5) Material Cost optimization.
- 6) Outer case design for overall protection.
- 7) Bus bar design.
- 8) Packaging constraint.

Why Thermal management imp.

- Temperature of the pack directly impacts
- 1) Electrochemical Reaction
 - 2) Efficiency of pack
 - 3) Charge acceptance
 - 4) Power & Energy availability.
 - 5) Safety & reliability.
 - 6) Life & life cycle cost

Required f's of Thermal Design

- 1) Compact
- 2) Light weight.
- 3) Easily packaged.
- 4) Reliable
- 5) Servicable
- 6) Low cost.
- 7) Low parasitic Power.

Electrical Design

Heat generation in battery pack (Joule heating)

Heat generation = Current (A)² × Internal resis (Ω)

→ Cells } heat generation. can be
→ Busbar } impacted by DOD, OCV,
→ BMS } Temperature & depends on
internal resis

* Always anticipate heat generated throughout life cycle
2P16S prismatic cell. * it doubles at end of life
15Ah, 3.65V nominal V, 10mΩ internal impedance.

If module is discharged from 100% to 20% SOC @ 1C.

Calculate heat generated & energy expelled by module.

$$\Rightarrow \text{heat generated } q_{\text{cell}} (1^2 r) = 2.25 \text{ W}$$

Heat generated by pack

Total cells, $n = 32$

Heat generated $q_{\text{pack}} (n * q_{\text{cell}})$

$$= 72 \text{ W}$$

Consider linear even though it is not.

Energy lost by pack as heat

Time for 100% to 20% SOC @ 1C, t
 $= 2880 \text{ s}$.

Heat energy lost $E_{\text{cell}} (q_{\text{cell}} * t) = 6.48 \text{ J}$

Heat lost by pack = $(n * E_{\text{cell}})$

$$= 207.36 \text{ KJ}$$

5045.76 → 207.36

100

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Comparison with total pack energy

Pack energy from 100% to 20% 30C

$$(n \times V \times I \times t) = 5045.76 \text{ kJ}$$

% energy lost as heat = 4/1%

if I save this my range goes up by approx 4%.

Environment aspect not considered. { we want 15-35 but if outside is 40°C

Then that has to be considered too.

→ Reducing Crate would also reduce heat energy lost.

→ Consider cells with low impedance / resistance

1 → For heat flow you need temp. gradient. just like for current to flow you need potential difference.

2 → Thermal resis. can be seen in all 3 modes of heat transfer

Conduction - $\frac{1}{kA} \Delta t$

Convection - $\frac{1}{hA} \Delta t$

Radiation -

Temp. gradient = $Q = \frac{T_1 - T_2}{R_{T \text{ and } R_{T \text{ cond.}}}}$

Electric current = $I = \frac{V_1 - V_2}{R}$

Modes of Thermal Management.

active

(need immediate energy to take out energy)

air, liquid, refrigerant, hybrid
(water + glycol)

Passive

(do not need)

PCM

Embedded Micro channels

For previous example

→ Identity thermal management system to maintain 45°C while ambient is 35°C .

Pack temp at

Convective area is 0.168 m^2

Heat generation rate of pack = 72 W
 72 J/s.

→ Thermal resis. $R_{th} = \frac{\Delta T}{Q}$

T_{cell}

$T_{ambient}$

Heat generated @

$\Delta T = T_{cell} - T_{ambient}$

$= 45^{\circ}\text{C} - 35^{\circ}\text{C}$

$= 10^{\circ}\text{C}$

Thermal resis = $R_{th} = 0.13 \text{ K/watt}$ ^{kelvin}

Pack temp

0.13 K/w

ambient

35°C

45°C

Convective $R_{th} = \frac{1}{hA}$

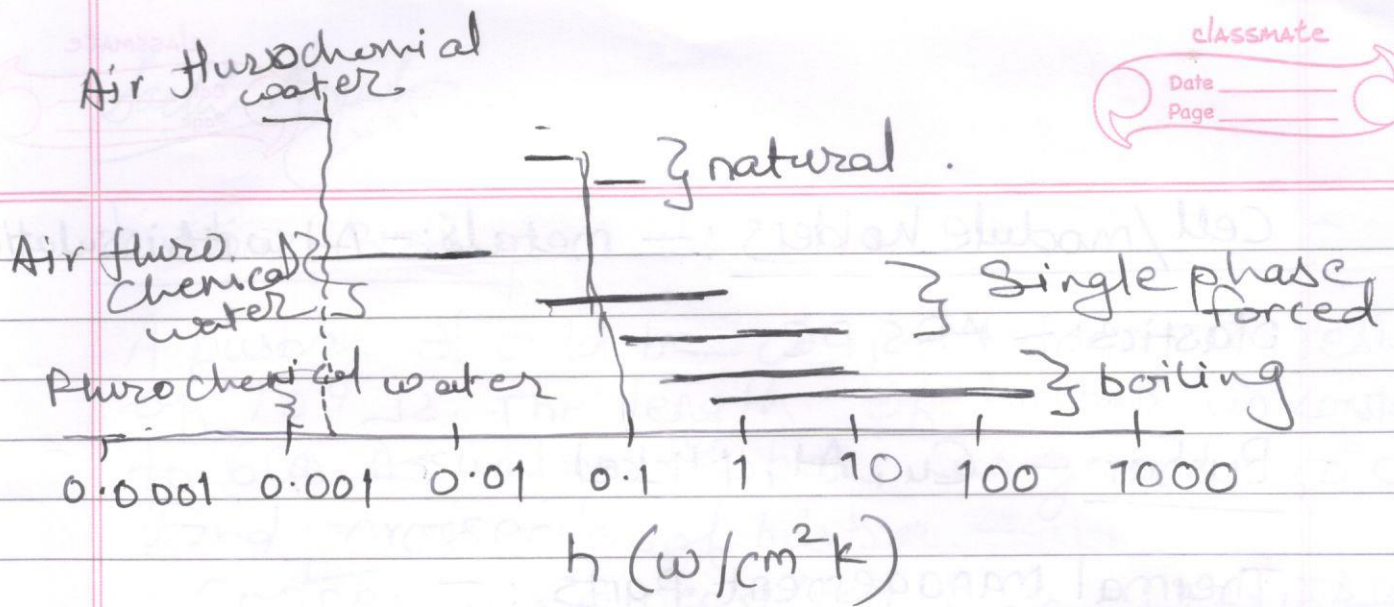
$\downarrow$
 0.13 K/w

Convective area = 0.168 m^2

convective $hTC = 0.0045 \text{ W/cm}^2\text{K}$

heat transfer coefficient

forced convection could work



★ Consider 2P16S pack with dimensions shown.

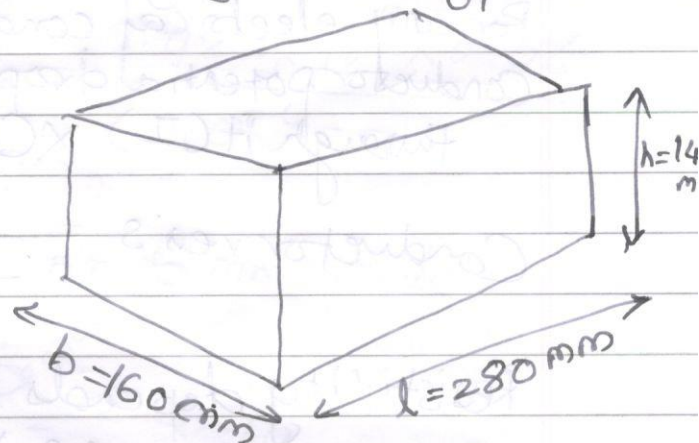
The pack is present at $45^\circ C$. Consider 2C discharge & identify suitable thermal management method to limit the cell temperature to $50^\circ C$. Assume heat is extracted only using bottom surface of pack.

Cell specifications

Voltage: 3.6V

Capacity: 15Ah

Internal Resis: $10 m\Omega$



$$Q = hA\Delta T$$

Passive: - Heat sink: - Increase the surface area
Usually crates are low. $0.05 W/cm^2$

Heat pipe cooling: - $5 W/cm^2$. Cu pipes

Used to decrease contact resis: - By using thermal pastes.

External casing: - Metals - Al, Steel
Plastics - ABS, PC
Composites - CFRP, Aramid-kewlar.

Cell/module holders : — metals: — Al with insulation

plastics: — ABS, PC

Busbar — Cu, Al, Nickel

Thermal management Parts : —

Cold plate — Al

Heat pipe — Cu

Coolant tubes — Al, ABS, EPDM.

Electrical Design

For any electrical conductor

Conductor potential drop (V) = Current flowing through it (I) × Conductor Resistance (R)

$$\text{Conductor res's} = \frac{\rho (\text{resistivity}) \times L (\text{conductor length})}{A (\text{conductor cross section area})}$$

Resistivity depends on temp.

$$\rho(T) = (\rho_{T_0}) [1 + \underbrace{\alpha(T - T_0)}_{\text{temp.}}]$$

reference temp.

Current carrying capacity of a busbar is limited by its max. acceptable temp.

heat generated by Joule heating <

heat dissipated to ambient

$$I^2 \cdot R_s \leq h A (T_{\text{busbar}} - T_{\text{ambient}})$$

R_s shape & proximity correction factor

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busbar sizing

A busbar should be designed for a max resis of $10^{-4} \Omega$. The length of busbar is constrained to 6cm. Assuming busbar can go upto 50°C . Find crosssection of busbar.

Copper: $\alpha = 0.000393^\circ\text{C}^{-1}$, $\rho_{20^\circ\text{C}} = 1.72 \times 10^{-8} \Omega \cdot \text{m}$

$\Rightarrow$

$$\rho_{50^\circ\text{C}} = \rho_{20^\circ\text{C}} (1 + \alpha \cdot (T - T_{\text{amb}}))$$

$$= (1.72 \times 10^{-8}) (1 + (0.000393)(50 - 20))$$

$$= 1.922 \times 10^{-8} \Omega \cdot \text{m}$$

The resis. of busbar $R = 10^{-4} \Omega$

$$R = \frac{\rho \cdot l}{A} \Rightarrow A = \frac{\rho \cdot l}{R} = \frac{1.922 \times 10^{-8} \times 0.06}{10^{-4}}$$

$$= 1.1 \times 10^{-5} \text{m}^2$$

Assuming busbar to be 5mm wide the thickness

$$t = \frac{1.1 \times 10^{-5} \text{m}^2}{0.005 \text{m}} = 2.2 \times 10^{-3} \text{m}$$

or 2.2 mm

★ Heat generation due to S.C. $\rightarrow$ higher temp
The rate of temp. rise during S.C. scenario can be obtained as

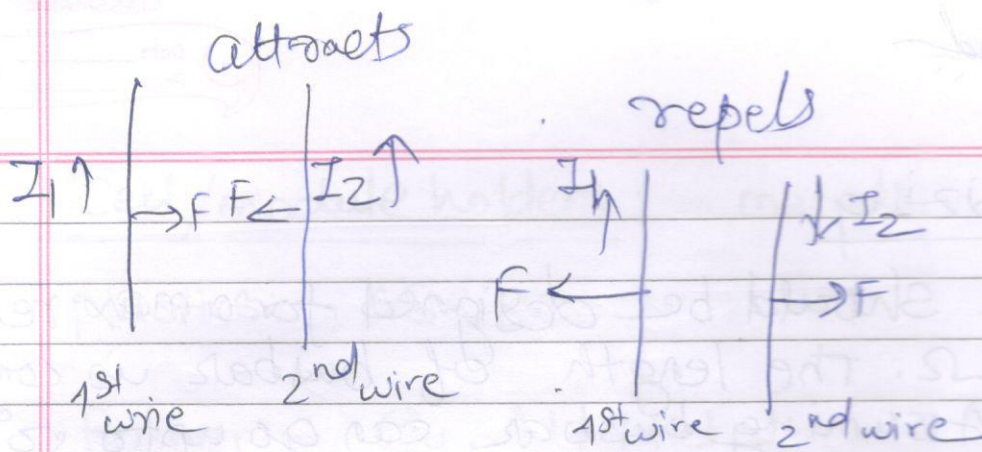
$$\Delta T_{\text{rise}} = \frac{\text{Resistivity}}{C_p \cdot \text{Density}} \left(\frac{I}{A} \right)^2$$

heat capacity of material

current density

temp $\uparrow$ resis $\uparrow$

Cu, $\rho \uparrow$ by 60%
as temp $\uparrow$ 20 to 300°C



electro magnetic force betⁿ
2 // ^l conductors.

emf betⁿ 2 // ^l current carrying
conductors induce force on each other
if current is high / consider
attraction & repel
forces.

Conductor Insulation

operating voltage - required dielectric
strength of insulation.

Operating temp - Upper & lower temp.
Environment

Material Potential to moisture absorption
- can affect creepage distance.

~~Problem~~

2P/6S config. for 1C discharge cell capacity
15 Ah.

Design a busbar for following specifications

- 1) Material Cu
- 2) $T_{\text{ambient}} = 30^{\circ}\text{C}$
- 3) Surface temp = ~~50~~ $^{\circ}\text{C}$
- 4) Convective heat transfer of 0.038 W
- Also calculate net voltage drop for 2P/6S

⇒ Heat generated by Joule effect $\leq$ ambient Heat dissipation

$$Q_{\text{joule}} = Q_{\text{conv}} \rightarrow I^2 R$$

$$= 0.038 \text{ W}$$

$$\text{Or } R = \frac{0.038}{900}$$

$$= 4.23 \times 10^{-5} \Omega$$

- Busbar thickness

$$t = \frac{1.885 \times 10^{-8} \times 0.059}{0.0065 \times 4.23 \times 10^{-5}}$$

$$= 3.98 \times 10^{-3} \text{ m or } 4 \text{ mm}$$

4mm bus bar is unsuitable for welding
so choose 7mm bus bar

$$\text{Resulting width} = \frac{26 \text{ mm}^2}{1 \text{ mm}} = 26 \text{ mm}$$

The resis of Cu busbar

$$R = \frac{\rho_{30^\circ\text{C}} \times L}{A}$$

$$= \frac{1.855 \times 10^{-8} \times 0.059 \text{ or } 0.06}{0.0065 \times 0.004}$$

$$= 4.2 \times 10^{-5} \Omega$$

Voltage drop across Busbar

$$V = IR$$

$$= (15 \times 2) \times 4.2 \times 10^{-5}$$

$$= 0.0012 \text{ V}$$

No. of 2P busbars in 2P/16 Spark = 15

$$\therefore \text{voltage drop} = 15 \times 0.0012 \text{ V}$$

Contact resis

- 1) Streamline resis $\rightarrow$ ^{Cleanliness} Smoothness of busbar
- 2) Contact resis $\rightarrow$ This is unavoidable & avoidable

every cell has thermistor & voltage sensor

* Voltage drop in sensor Harness

- 1) Harness provides electrical pathway
- 2) Harness being electrical conductor, voltage drop across every ends
- 3) Critical when harness measures at both nearer & farthest places
- 4) Longer conductor $\rightarrow$ higher voltage drop & hence incorrect measurement

Sensing system is calibrated using:

1. Software (multiplier & gain factor)
2. The harness (equal resis. in all wires to



Consider

wire gauge / diameter = 20 AWG / 0.81 mm
length = 300 mm

The corresponding wire resis. would be
0.01 Ω for Cu.

$$\text{Voltage drop} = 0.02 \times 0.01$$

$$= 0.0002 \text{ V small}$$

but in series it counts.

Thicker wire for
small wire

thickness
adjust to
keep

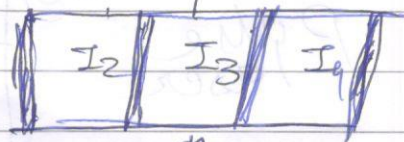
res. equal

Current equalization in || path

if current is not same.

I_1 & I_4 res is more

I_2 & I_3 has ~~high~~ ^{small} resis.



current flow

But it is designed for same current

but I_2 & I_3 will carry more current &
mosfet will be out of fail

then current flows through I_1 & I_4 but
now all current flows through them &
increases temp. & fails it.

Provide higher cross section to longer paths
helps to achieve same resis. to that of a
shorter path

$$\text{path resis} = R = \frac{\rho l}{A}$$

vary A based on l
to get same res.
to get same current

By equal resis in all paths $R_1 = R_2 = R_3 = R_4$

$$= \frac{l_1}{A_1} = \frac{l_2}{A_2} = \frac{l_3}{A_3} = \frac{l_4}{A_4} = \frac{l_{avg}}{A_{avg}}$$

Busbar Joining

Resis welding	→ Fast → low cost → easy automation → controlled	Difficult for highly conductive materials & Can be used for small points
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Laser welding Continuous or Pulse laser	→ High speed → easy → less thermal → non contact process (because harmful lasers)	automation required
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★ BMS ★

- Controls in & out of current
- when crash occurs it makes sure everything is safe

Importance

- 1) Safety → Controls the functional limit of part
 → Electrical & thermal.
- 2) Lifecycle/cost

3) Measurement & Prediction

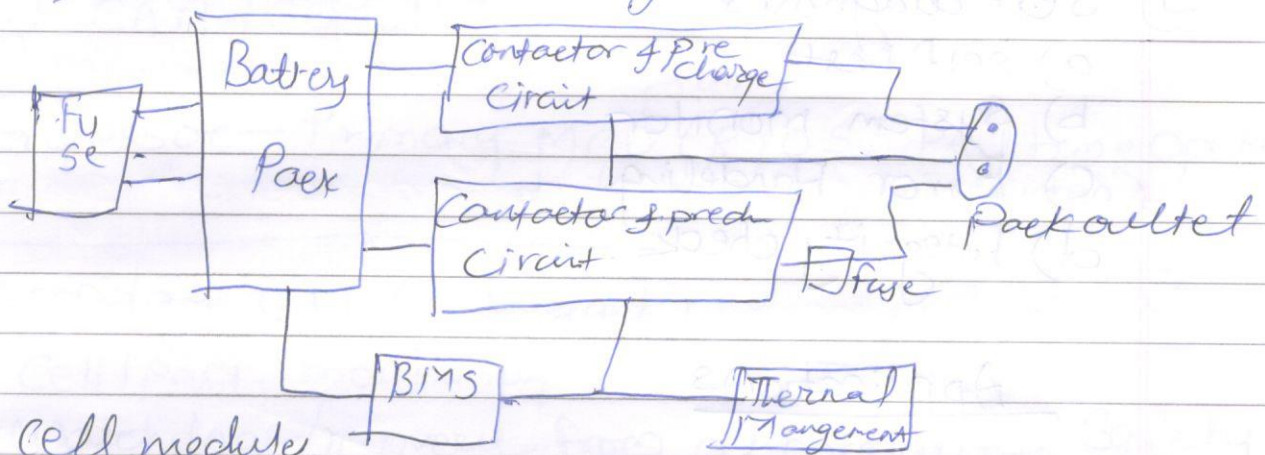
SoC & SoH



How much more cycles it can go through
how much more you can drive

- 4) Abuse :- If someone thunders or draws more current it records such events.
logs temps & change in ambient temp etc.

Battery pack layout



1) Cell module

2) BMS

3) Insulation management device → to separate LT & HT

4) Precharge Circuit

5) Current sensors

6) Fuse

7) Balancing unit

& send signal to BMS to cut off

Core BMS fn

1) Safe operation

- a) Temp
- b) Volt
- c) Current

2) Database

- a) Cell parameters
- b) Balancing status
- c) I/O requests

3) Self diagnosis

- a) self test
- b) System monitor
- c) Error Handling
- d) Integrity check

Applications

1) State estimation

- a) SOC
- b) SOH

2) Advanced applications

- a) State modelling :- when hard to measure we predict or model
- b) Prediction/estimation

Data Storage

- 1) SDRAM 2) EEPROM

Battery system ctrl

- contactor control
- Thermal control.
- .

Interface drivers

- cell monitoring
- Isolation monitoring
- CAN / Serial.

Processor - Primary MCU (RTOS - Real time operating system) ^{unit}
_{Master control}

Secondary CU (redundant)

- 1) Cell/pack monitoring
 - 2) Watchdog / recovery from abnormality
 - 3) Safe operating area.
- Safety

- 1) Serial communication
- 2) Interlock control

Interface device

Battery Pack Sensing

Voltage

- Individual cell voltage
- Critical i/p to SOC & SOH parameters estimation
- Essential for protection — higher or lower voltage
- Measured using IC's

Temperature; —

- Short term cell performance prediction
- depend on cell temp.
- Cell degradation mechanism are also temp. dependent.
- Critical i/p for thermal management sys. function
- Mostly measured using thermistors.

Current; —

- Used to detect & log abuse conditions & ensure safety
- Critical i/p to SOC & SOH
- Estimated mainly using current shunts and hall effect sensors.

Isolation Sensing

- All wiring fully insulated → HV battery terminal not connected to any of LV system or highly exposed vehicle chassis → Enhances the safety & accident prevention.

Sensing Voltage Signals (ADC)

- Voltage signal is analog
 → Microcontroller reads digital
 → ADC converts voltage/current signal to MC readable formation.
 → Higher ADC bit rating, we high frequency discrete high resolution, higher bit rating.
 10 bit ADC = $2^{10} = 1024$ level.

1024

→ analog
 → digital

for 5 bit it will be $2^5 = 32$

Sampling rate
 Max. resolution



Relating ADC value to voltage.

- ADC reports ratiometric values.

- $$\frac{\text{ADC Resolution}}{\text{System voltage}} = \frac{\text{ADC reading}}{\text{Measured analog voltage}}$$
- Max. cell voltage 4.5V, 10 bit ADC
 If cell is at 3.6V

$$\text{then ADC} = \frac{1024}{(4.5-0)} = \frac{\text{ADC reading}}{3.6}$$

$$\text{ADC would read out} = \frac{3.6 \times 1024}{4.5} = 818.$$

Question :- An 8 bit ADC used for 5V voltage sensor.

If ADC feeds 200 to μP what is the voltage sensed.

Sensing Thermistor values

$$i = \frac{V}{R_1 + R_{\text{Thermistor}}}$$

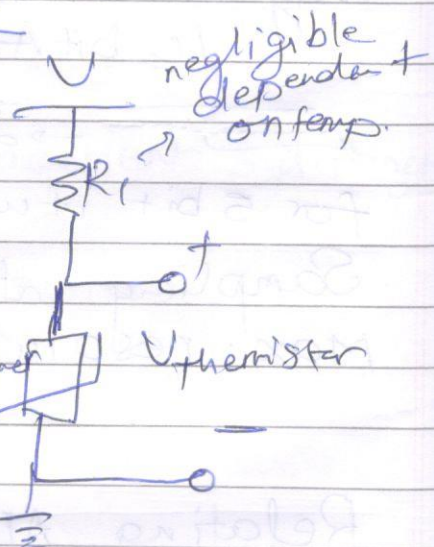
Measured thermistor voltage (using ADC)

$$V_{\text{Thermistor}} = \frac{R_{\text{Thermistor}}}{R_1 + R_{\text{Thermistor}}} \times V$$

Voltage divider

$$R_{\text{Thermistor}} = \frac{V_{\text{Thermistor}}}{V_1 + V_{\text{Thermistor}}} \times R_1$$

V_1 = Source voltage



Example

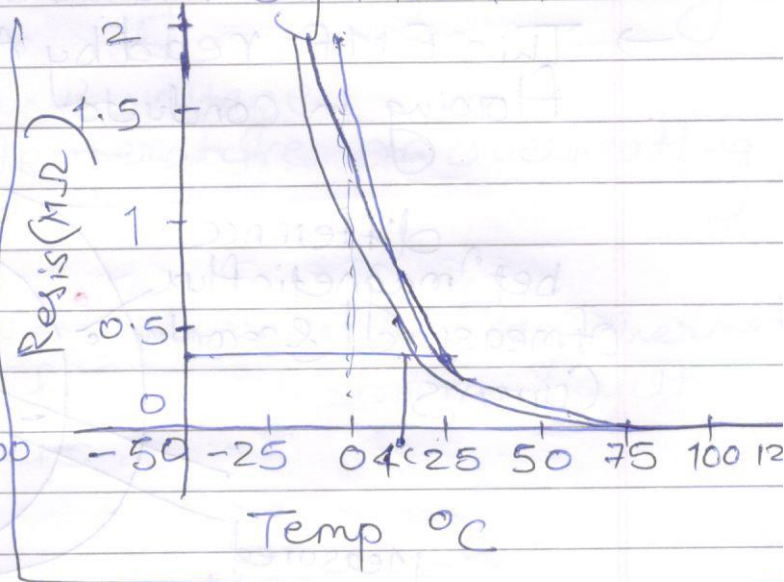
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If a thermistor is placed in lower leg of voltage divider circuit of 5V source & $R_1 = 100 \text{ k}\Omega$. The voltage across thermistor is 2V. Using plot shown find temp. sensed by Thermistor

$$R_{\text{Thermistor}} = \frac{V_{\text{Thermistor}}}{V_1 + V_{\text{Thermistor}}} R_1$$

$$R_{\text{Thermistor}} = \frac{2}{5+2} \times 100 \times 1000$$
$$= 0.28 \text{ M}\Omega$$



From plot, we roughly get Temperature to be 4°C .
we feed this table to μP

Current Sensing.

Current sensing is done in 2 ways

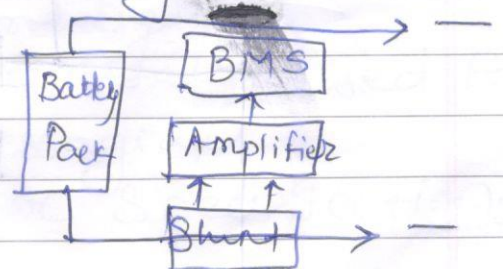
1. Current shunt

→ Current shunt uses low values (like $0.1 \text{ m}\Omega$) high precision resistor in series

→ lowest resistance aid in lower Joule heating.

→ By sensing Voltage across shunt using ADC, Current can be calculated using ohm's law.

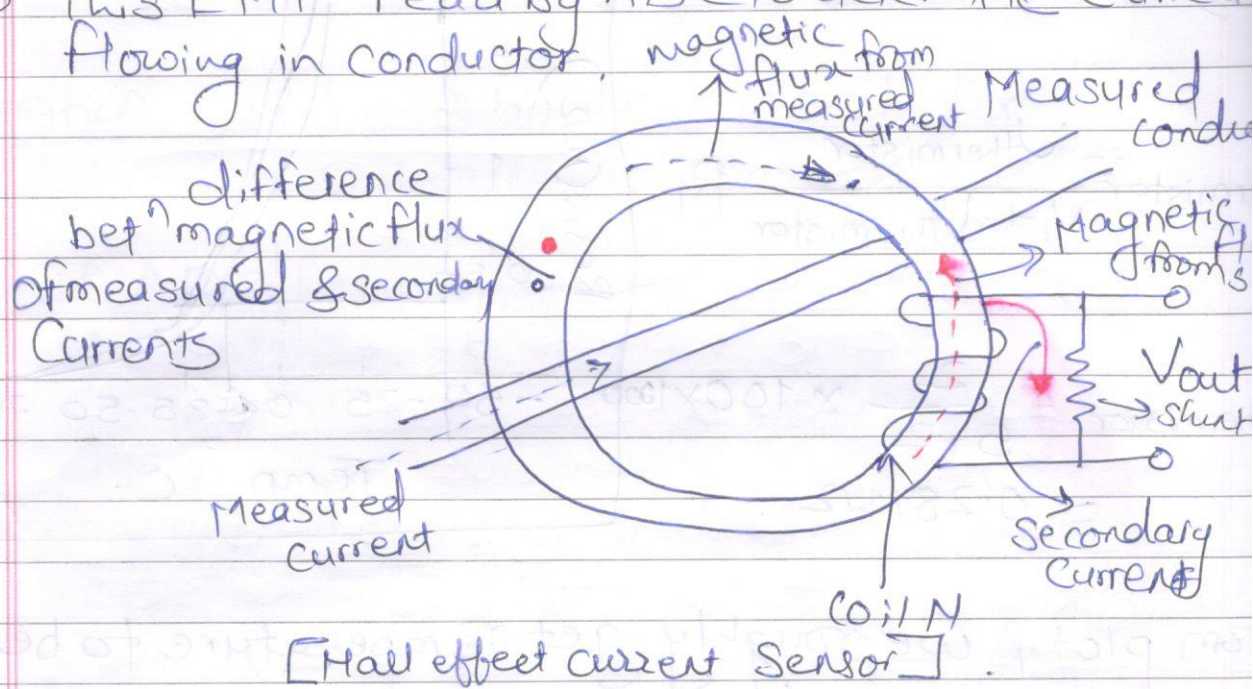
$$I = \frac{V_{\text{shunt}}}{R_{\text{shunt}}}$$



2. Hall effect sensor.

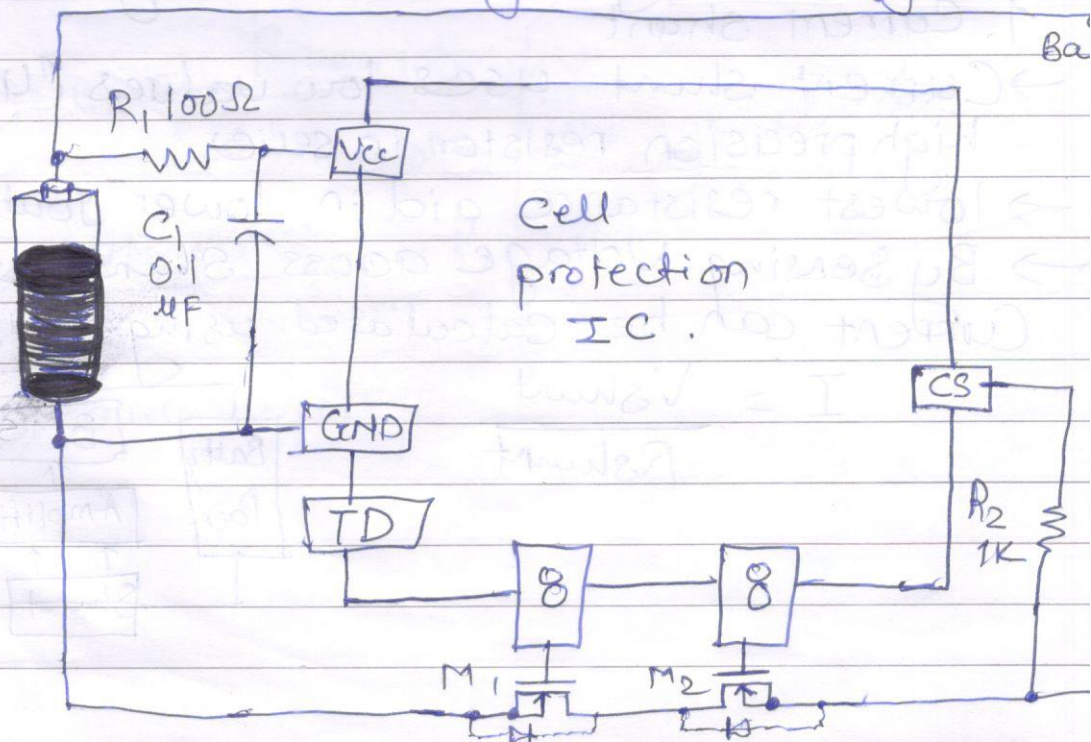
→ Current flowing through conductive coil, induces EMF in it.

→ This EMF read by ADC to determine current flowing in conductor.



Battery Pack Protection :-

BMS software & electronics are integral parts of an overall management strategy.



Following undesirable events/conditions are addressed

- Excessive current during charging or Discharging
- short ckt
- Overvoltage or undervoltage
- High ambient temperature or overheating
- Loss of Isolation
- Abuse

Protective/Protection devices include Thermal fuses, conventional fuses, Electronic fault detection like IMD etc.

Battery pack interfacing

Effective communications interfacing betⁿ BMS & the application being powered by the battery pack. These include

- charger control
 - Must be capable of handling random charges (of course in limit)
 - Must be adaptable to different rate of charging depending on the electric utilities grid.
- Communication protocols determine the transmission speed, priority settings, transmission sequence, error detection, & handling and control signals.
- ★ CAN is almost universally used for on board vehicle messaging
- ★ CAN has an electric Specification for packet control.

→ Logbook Function

- For warranty & diagnostic purpose, BMS must store a log of typical & abuse events.
- Mostly stored in a flash memory.

Communication

There are 2 communication pathways for BMS:

1. Betⁿ Master & Slave BMS
2. Betⁿ Master BMS & vehicle ECUs.

They can also be wireless like Bluetooth where distance is less.

There are many communication protocols which BMS use for communication.

1. CAN bus (more common)
2. SPI bus (ps 485 etc)
3. I2C bus.

CAN interface

→ stands for Controller area Network

Most robust protocol in wide use.

→ Benefits over other protocols

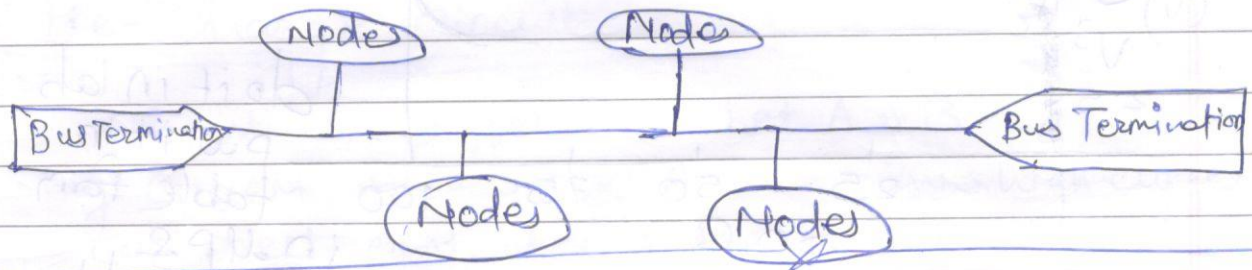
→ Simple & low cost

→ Fully centralized

→ Extremely robust

→ Efficient.

Each ECU is called as Node, betⁿ which communication occurs.



★ Battery Pack Performance Management ★

Soc, soh, voltage are performance indicator.

If I want 30A but contactor gives 20A. performance gets limited to 20A.

So software & hardware should support whatever is the battery capacity.

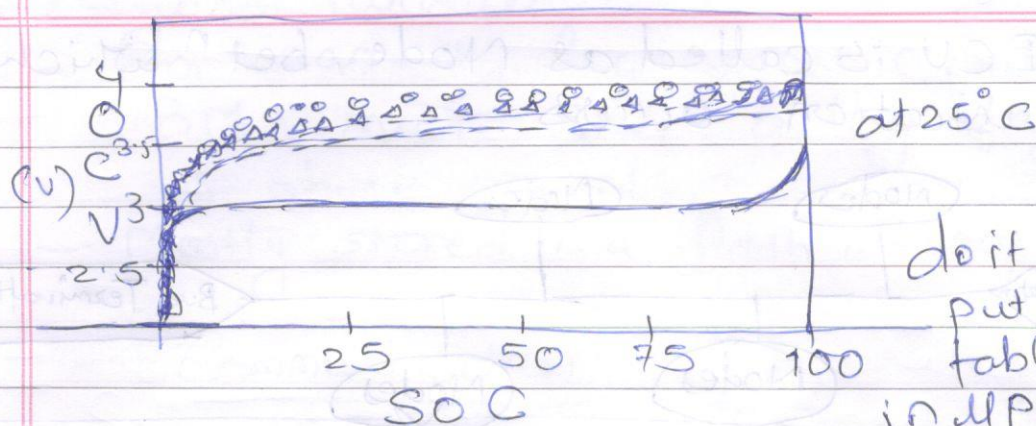
for example if battery is asked to give 40A & can give 40A then software should allow it & hardware should be able to cope up with it.

In order to perform ~~reliability~~ reliably, the application must have regular updates on the performance parameters like available energy & power.

Thus, BMS must report following performance parameters:

State of charge:-

- Depends on average particle concentrations on electrodes
- estimated using current, coulombic efficiency & cell total capacity.



do it in lab
put it in
table form
in uP &
can be used to
determine SOC

Energy available

- Ability to do work measured in Wh estimated as a function of OCV & state of Charge.
- Critical I/P for measuring range of EV.

Power estimate

- Rate at which energy can be moved from batteries measured in W
- Mostly stored in flash memory.

Battery-Pack Diagnostics

Keep record of all atypical events.
helps to do troubleshooting.

does it → Logbook : - keeps count of no. of cycles of charge/discharge.

- Contains details of any atypical event.

State of Health

- Cell model internal parameters like cell present total Capacity & series resistance
- Capacity fade : Capacity decreases in range of 20% to 30%.

Power fade : - Resis. generally increases in range of 50% to 100%.

★ Pre-Charge Circuit

Potential difference bet A & B
high current flows from A to B. This high current can melt contactor & then it gets welded.

★ Dangerous failure mode in an EV in which the contactor gets welded shut. CBMs cannot understand that)

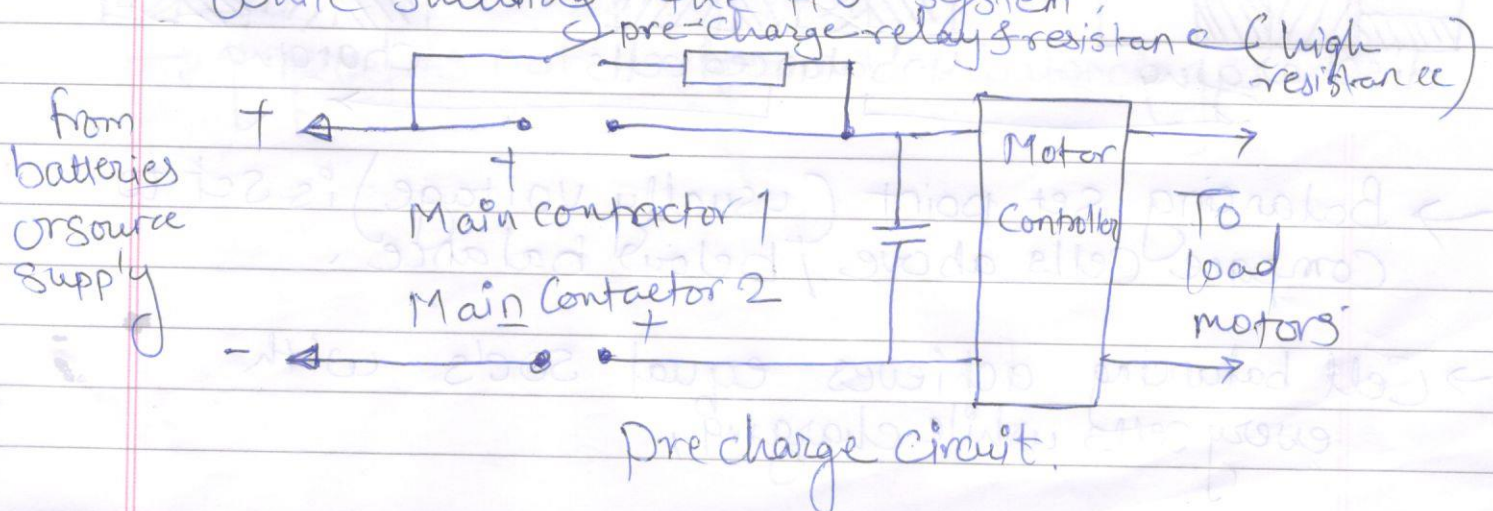
Reduce the potential circuit.
It is called pre charge circuit.

So it is used initially & then removed.

★ Contactors are basically relays present in high current path

★ If they are welded shut, the high voltage system will remain alive even after contactor is de-energized.

★ Pre charge circuit is 11k path to contactors which energize at least 90% of peak voltage before energizing the contactor. Same followed while shutting the HV system.



TI → Texas Instrument

Management Controller Unit MC

- MCU is brain of BMS, ITPs in form of a MC
- state Determination
- Data acquisition
- Pack control - Electrical & Thermal (both at cell & pack level)
- Safety protection - work with electrical / thermal fuse
- Communication Imp etc.

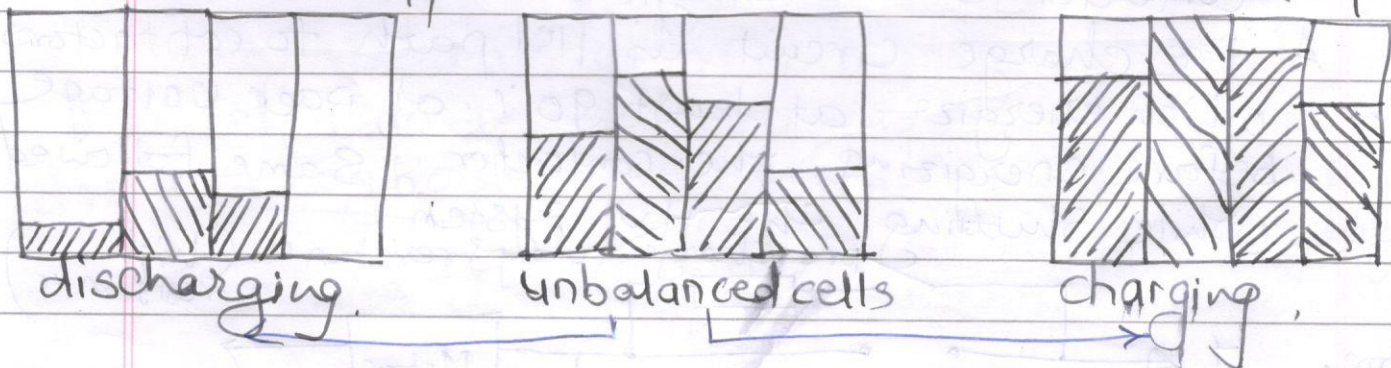
Primary & secondary MCU for redundancy.

★ Cell balancing ★

Degrade the pack.

Not every cell operates in similar way, so it can create imbalance in charging or discharging. because it does not charge/discharge uniformly.

→ Essential to achieve uniform charge/discharge to avoid wasted energy / wasted capacity.



→ Balancing set point (usually voltage) is set to compare cells above / below balance.

→ Cell balancing achieves equal SOC's with every cells while charging,

1. Non-Dissipative balancing, also active
2. Dissipative Balancing, also passive.

Reasons of imbalancing

- 1) Different coulombic efficiency of cells
- 2) Different charging current for different cells based on Internal Resis.
- 3) If busbar thickness is varied & has different resistance in different areas.
- 4) Temperature gradient is different for ~~both~~ ^{different} cells.

Cell balancing - Active & Passive.

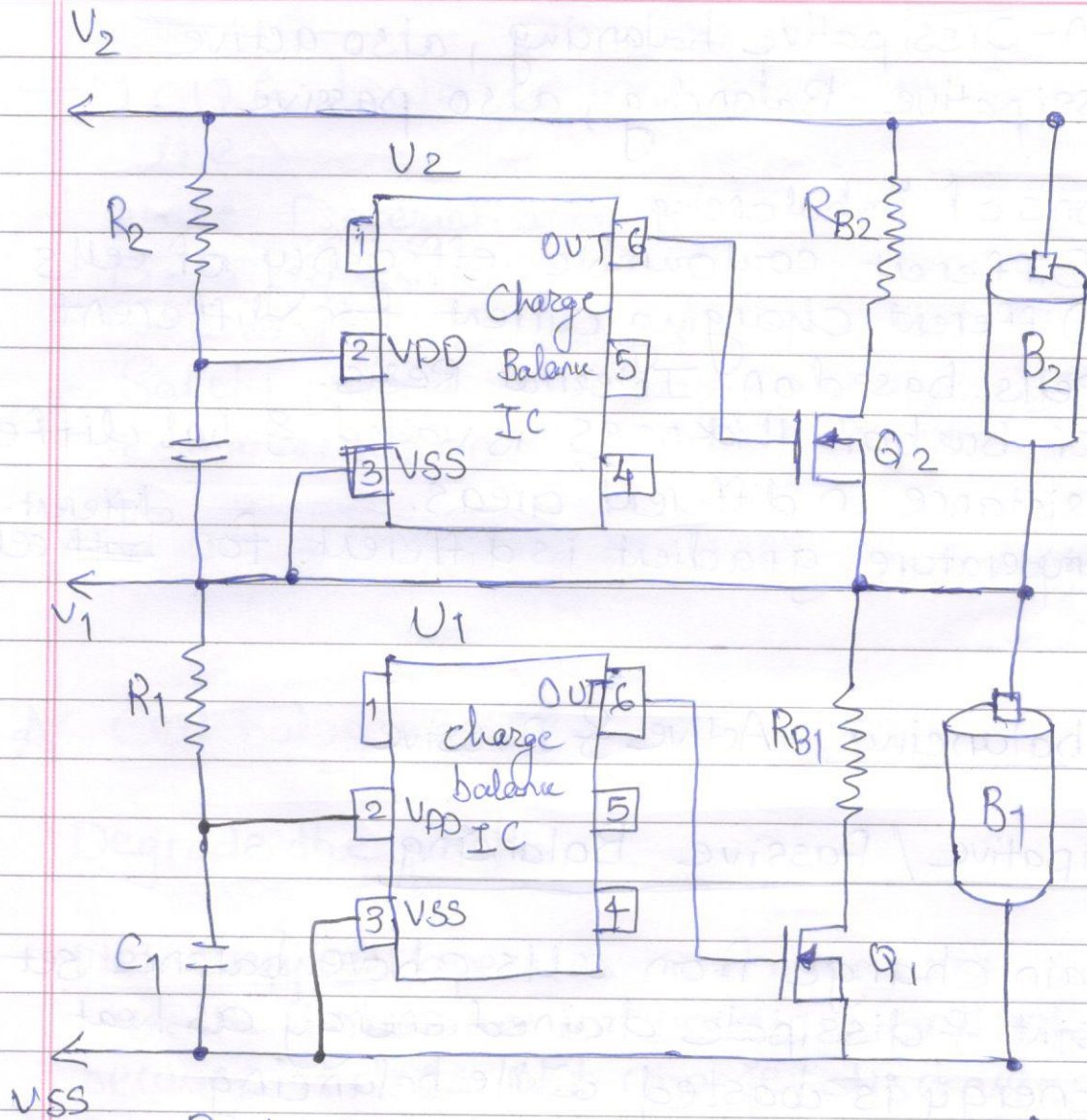
Dissipative / Passive Balancing.

- Drain charge from cells above balance set point & dissipate drained energy as heat.
- Energy is wasted while balancing.

pack life

Non-Dissipative / Active balancing.

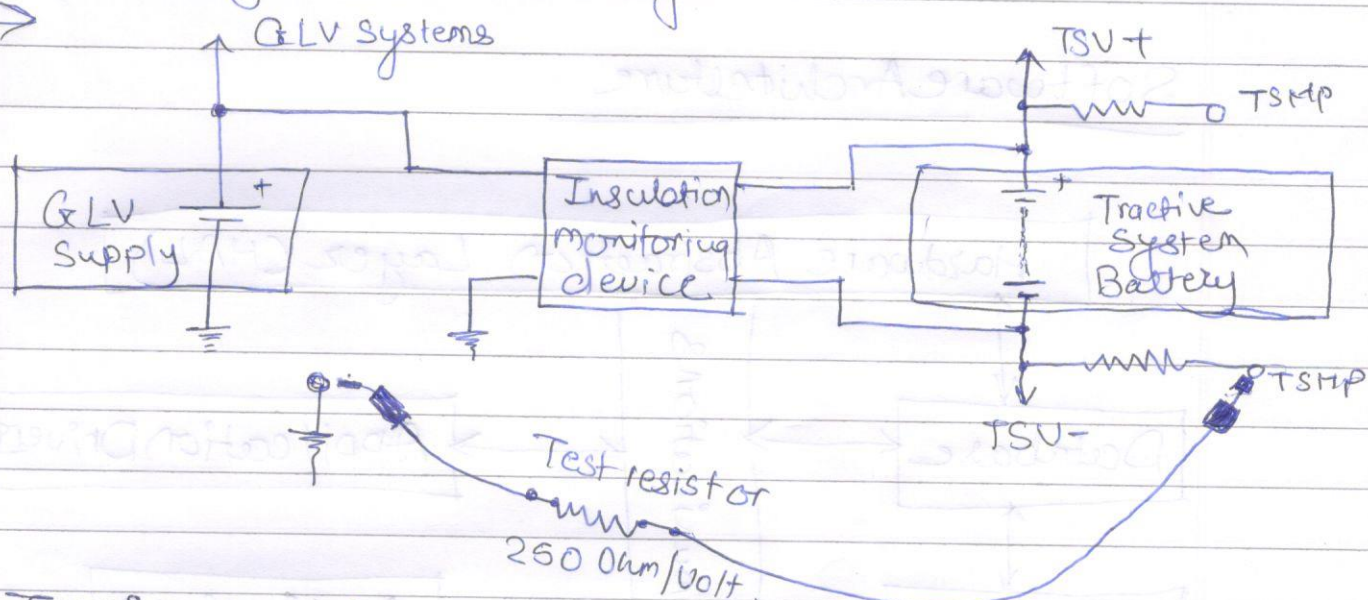
- Move charge from cells above balancing points to cells below balance point.
- Energy is not wasted while balancing.



Battery Charge balance IC Application
Hycon Tech - HY22/21C

Insulation Monitoring Device (IMD)

- Active safety device
- Present betⁿ Hot conductors & Ground.
- Detect any unintentional leakage path betⁿ HV & LV Circuits
- When leakage is detected, the IMD immediately shuts off the HV circuit & throws a non-resetable Error, which must be addressed during the servicing.



Testing IMD functionality by shorting HV to GND.

- Benefits
 - 1) Increased reliability
 - 2) Accident Prevention
 - 3) Active HV/LV monitoring
 - 4) Avoid costly overhauls.

1. Club to gether all the cells mps
2. Series Fuse
3. Contactor through BM to Contactor ^{energy}
4. Pre charge ckt through connector
5. or if you use mosfet ^{as switching ckt} through BM
6. Diff

1. Cell modules

2. Module series fuse

Software Architecture

